



Gamma knife radiosurgery for uveal melanomas and metastases: a systematic review and meta-analysis

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Summary

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Background Gamma knife radiosurgery is regarded as the gold-standard stereotactic radiosurgery modality for the treatment of intracranial tumours, and its use has been expanded for the treatment of intraocular malignancies. The aim of this study was to systematically evaluate the efficacy, outcomes, and complications of gamma knife radiosurgery for uveal melanomas and metastases.

Methods We did a systematic review and meta-analysis to aggregate the clinical outcomes of patients with uveal melanomas or intraocular metastases treated primarily with gamma knife radiosurgery. We searched MEDLINE and Embase for studies published between Sept 1, 1960, and Feb 1, 2020, reporting the use of gamma knife radiosurgery as primary treatment for uveal melanoma or uveal metastases. The search was restricted to clinical studies and relevant grey literature published in English. Studies reporting treatment of benign tumours, extraocular tumours, or other forms of stereotactic radiosurgery were excluded to reduce heterogeneity. No restrictions were placed on participant criteria. Local tumour control and tumour regression were extracted as the primary outcomes and analysed via a random-effects meta-analysis of proportions using the DerSimonian and Laird method with a Freeman-Tukey double arcsine transformation. This study is registered with PROSPERO, CRD42019148165.

Findings Our search returned 454 studies, of which 109 were assessed for full-text eligibility. 52 studies, reporting on 1010 patients with uveal melanoma and 34 intraocular metastases, were eligible for systematic review. 28 studies were included in the meta-analysis. 840 of 898 patients (0·96, 95% CI 0·94–0·97; $P=16\%$) from 19 studies had local control, and 378 of 478 patients (0·81, 0·70–0·90; $P=83\%$) from 16 studies experienced tumour regression.

Interpretation Gamma knife radiosurgery is an efficacious primary method of treating uveal melanomas and intraocular metastases, with reliable tumour control rates. Randomised controlled trials should further evaluate the safety and efficacy of gamma knife radiosurgery in this setting.

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Introduction

Uveal melanomas are the most common primary intraocular tumour in adults,¹ with choroidal metastases from distant primaries, such as the breast, lung, or gastrointestinal tract, being the most common metastases associated with these intraocular tumours.² Before the Collaborative Ocular Melanoma Study (COMS), the standard of care involved radical surgery, including enucleation or orbital exenteration, to prolong survival. However, the randomised, multicentre COMS clinical trial,³ completed in 2003, showed that iodine-125 brachytherapy has a similar survival rate for medium-sized uveal melanomas compared with enucleation, resulting in a shift towards globe-preserving therapies to treat intraocular tumours.

Although brachytherapy remains the primary modality of treatment, various forms of stereotactic radiotherapy have shown similar efficacy in controlling tumour burden⁴ and can additionally be used to treat circumpapillary or large uveal melanomas that might not be amenable to plaque brachytherapy.

To date, many primary studies have been published on the efficacy of gamma knife stereotactic radiosurgery in treating uveal tumours but, to our knowledge, no meta-analyses have been done to systematically evaluate the quality, outcomes, and complications of this intervention. Gamma knife radiosurgery is widely considered to be the benchmark of stereotactic radiosurgery⁵ and remains the most widely used form of stereotactic radiosurgery across the USA,⁶ due to its precision and the efficiency of radiation delivered in a single session. The aim of this systematic review was to aggregate the current literature regarding the treatment of uveal melanomas and metastases using gamma knife radiosurgery as the primary therapy and evaluate its efficacy on tumour control and regression. We also aimed to evaluate visual preservation, secondary enucleations, radiation-induced complications, and overall survival after gamma knife radiosurgery.

Methods

Search strategy and selection criteria

We did a systematic review and meta-analysis to evaluate the clinical outcomes of gamma knife radiosurgery when